

Your Paper

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Abstract

Your abstract.

1 Introduction

2 Implementation of MHD equations

2.1 Normalized Equations

$$\frac{\partial n_i}{\partial t} + \nabla \cdot (n_i \mathbf{V}) = 0 \quad (1)$$

$$\frac{\partial n_n}{\partial t} + \nabla \cdot (n_n \mathbf{U}) = 0 \quad (2)$$

$$\frac{\partial n_i \mathbf{V}}{\partial t} = -\nabla \cdot \left[n_i \mathbf{V} \mathbf{V} + n_i T_i + \frac{B^2}{2} \mathbf{I} - \mathbf{B} \mathbf{B} \right] + n_i \mathbf{g} - n_i \nu_{in} (\mathbf{V} - \mathbf{U}) \quad (3)$$

$$\frac{\partial n_n \mathbf{U}}{\partial t} = -\nabla \cdot \left[n_n \mathbf{U} \mathbf{U} + \frac{n_n}{m_n} T_n \right] + n_n \mathbf{g} + \frac{n_i \nu_{in}}{m_n} (\mathbf{V} - \mathbf{U}) \quad (4)$$

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \cdot (\mathbf{V} \mathbf{B} - \mathbf{B} \mathbf{V}) - \nabla \times \left(\eta \mathbf{J} + \frac{1}{n_e e} \mathbf{J} \times \mathbf{B} \right) \quad (5)$$

$$\begin{aligned} \frac{\partial T_i}{\partial t} = & -\nabla \cdot (T_i \mathbf{V}) + \frac{1}{3} T_i \nabla \cdot \mathbf{V} - \frac{2}{3} \frac{\nabla \cdot (\mathbf{q}_i + \eta \mathbf{B} \times \mathbf{J})}{n_i} \\ & + \frac{2}{3} \cdot \frac{1}{1 + m_n} \nu_{in} \left[3(T_n - T_i) + m_n |\mathbf{V} - \mathbf{U}|^2 \right], \end{aligned} \quad (6)$$

$$\begin{aligned} \frac{\partial T_n}{\partial t} = & -\nabla \cdot (T_n \mathbf{U}) + \frac{1}{3} T_n \nabla \cdot \mathbf{U} - \frac{2}{3} \frac{\nabla \cdot \mathbf{q}_n}{n_n} \\ & + \frac{2}{3} \cdot \frac{m_n}{1 + m_n} \nu_{in} \left[3(T_i - T_n) + |\mathbf{V} - \mathbf{U}|^2 \right], \end{aligned} \quad (7)$$

$$\nabla \times \mathbf{B} = \mathbf{J} \quad (8)$$

$$\mathbf{q} = -\kappa \nabla T. \quad (9)$$

2.2 General Lagrange Multiplier

Introducing a general Lagrange multiplier (Φ) , (5) becomes

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \cdot (\mathbf{V} \mathbf{B} - \mathbf{B} \mathbf{V}) - \nabla \times \left(\eta \mathbf{J} + \frac{1}{n_e e} \mathbf{J} \times \mathbf{B} \right) + \nabla \Phi, \quad (10)$$

with

$$\frac{\partial \Phi}{\partial t} + c_h^2 \nabla \cdot \mathbf{B} = -\frac{c_h^2}{c_p^2} \Phi. \quad (11)$$

2.3 Normalized Fluxes

1. Matrix Form of F_{ni} :

$$F_{ni} = \begin{pmatrix} n_i v_x \\ n_i v_y \\ n_i v_z \end{pmatrix}$$

2. Matrix Form of F_{nn} :

$$F_{nn} = \begin{pmatrix} n_n u_x \\ n_n u_y \\ n_n u_z \end{pmatrix}$$

3. Matrix Form of F_V :

$$F_V = \begin{pmatrix} \frac{b^2}{2} - b_x^2 + n_i T_i + n_i v_x^2 & -b_x b_y + n_i v_x v_y & -b_x b_z + n_i v_x v_z \\ -b_x b_y + n_i v_x v_y & \frac{b^2}{2} - b_y^2 + n_i T_i + n_i v_y^2 & -b_y b_z + n_i v_y v_z \\ -b_x b_z + n_i v_x v_z & -b_y b_z + n_i v_y v_z & \frac{b^2}{2} - b_z^2 + n_i T_i + n_i v_z^2 \end{pmatrix}$$

4. Matrix Form of F_U :

$$F_U = \begin{pmatrix} \frac{n_n T_n}{m_n} + n_n u_x^2 & n_n u_x u_y & n_n u_x u_z \\ n_n u_x u_y & \frac{n_n T_n}{m_n} + n_n u_y^2 & n_n u_y u_z \\ n_n u_x u_z & n_n u_y u_z & \frac{n_n T_n}{m_n} + n_n u_z^2 \end{pmatrix}$$

5. Matrix Form of F_B :

$$F_B = \begin{pmatrix} -\Phi & b_y v_x - b_x v_y & b_z v_x - b_x v_z \\ -b_y v_x + b_x v_y & -\Phi & b_z v_y - b_y v_z \\ -b_z v_x + b_x v_z & -b_z v_y + b_y v_z & -\Phi \end{pmatrix}$$

6. Matrix Form of F_{Ti} :

$$F_{Ti} = \begin{pmatrix} T_i v_x \\ T_i v_y \\ T_i v_z \end{pmatrix}$$

7. Matrix Form of F_{Tn} :

$$F_{Tn} = \begin{pmatrix} T_n u_x \\ T_n u_y \\ T_n u_z \end{pmatrix}$$

8. Matrix Form of F_Φ :

$$F_\Phi = \begin{pmatrix} c_h^2 b_x \\ c_h^2 b_y \\ c_h^2 b_z \end{pmatrix}$$

2.4 Flux Derivative

The derivative of F is needed for upwind scheme. $G = \frac{d}{du_j} F_{ij}$ is calculated as below

$$G_V = \begin{pmatrix} 2 n_i v_x & n_i v_x & n_i v_x \\ n_i v_y & 2 n_i v_y & n_i v_y \\ n_i v_z & n_i v_z & 2 n_i v_z \end{pmatrix}$$

$$G_U = \begin{pmatrix} 2 n_n u_x & n_n u_x & n_n u_x \\ n_n u_y & 2 n_n u_y & n_n u_y \\ n_n u_z & n_n u_z & 2 n_n u_z \end{pmatrix}$$

$$G_B = \begin{pmatrix} 0 & v_x & v_x \\ v_y & 0 & v_y \\ v_z & v_z & 0 \end{pmatrix}$$

$$G_{ni} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}$$

$$G_{nn} = \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}$$

$$G_{Ti} = \begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix}$$

$$G_{Tn} = \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix}$$

2.5 Numerical Scheme

The function $F_x()$ calculates terms acted on by $\frac{\partial}{\partial x}$, which are $F_{xx}\hat{x} + F_{xy}\hat{y} + F_{xz}\hat{z}$.

Here for simplicity, I use the upwind scheme to obtain the numerical flux. I'll try the TVD-MUSCL scheme later.

The upwind scheme for nonlinear fluxes is

$$\hat{F}_{j+1/2} = \frac{1}{2} (f_j + f_{j+1}) - \frac{1}{2} |\hat{a}_{j+1/2}| (u_{j+1} - u_j). \quad (12)$$

where the speed $\hat{a}_{j+1/2}$ is defined as

$$\hat{a}_{j+1/2} = \begin{cases} \frac{f_{j+1} - f_j}{u_{j+1} - u_j} & \text{when } u_{j+1} \neq u_j, \\ f'(u_j) & \text{when } u_{j+1} = u_j. \end{cases} \quad (13)$$

2.6 TVD-MUSCL

The nonlinear numerical flux is handled in this form:

$$\begin{aligned} f_{j-1/2}^* &= \frac{1}{2} \left(\left[f(u_{j-1/2}^r) + f(u_{j-1/2}^l) \right] - \lambda_{j-1/2} [u_{j-1/2}^r - u_{j-1/2}^l] \right), \\ f_{j+1/2}^* &= \frac{1}{2} \left(\left[f(u_{j+1/2}^r) + f(u_{j+1/2}^l) \right] - \lambda_{j+1/2} [u_{j+1/2}^r - u_{j+1/2}^l] \right), \end{aligned} \quad (14)$$

where $\lambda_j = \max\{\rho[\frac{\partial f}{\partial u}(u_j^l)], \rho[\frac{\partial f}{\partial u}(u_j^r)]\}$ and $\rho[A]$ denotes the spectral radius of matrix A , and where

$$\begin{aligned} u_{j-1/2}^l &= u_{j-1} + \frac{1}{2} \phi(r_{j-1}) (u_j - u_{j-1}), \\ u_{j+1/2}^l &= u_j + \frac{1}{2} \phi(r_j) (u_{j+1} - u_j), \\ u_{j-1/2}^r &= u_j - \frac{1}{2} \phi(r_j) (u_{j+1} - u_j), \\ u_{j+1/2}^r &= u_{j+1} - \frac{1}{2} \phi(r_{j+1}) (u_{j+2} - u_{j+1}), \\ r_j &= \frac{u_j - u_{j-1}}{u_{j+1} - u_j}, \\ \phi(r) &= \max(0, \min(1, r)), \lim_{r \rightarrow \infty} \phi(r) = 1, \end{aligned} \quad (15)$$

Rewrite all unknowns in vector form:

$$\mathbf{u} = \begin{pmatrix} n_i \\ n_n \\ n_i v_x \\ n_i v_y \\ n_i v_z \\ n_n u_x \\ n_n u_y \\ n_n u_z \\ b_x \\ b_y \\ b_z \\ T_i \\ T_n \\ \Phi \end{pmatrix}, \mathbf{F}_x = \begin{pmatrix} n_i v_x \\ n_n u_x \\ \frac{b_z}{2} - b_x^2 + n_i T_i + n_i v_x^2 \\ -b_x b_y + n_i v_x v_y \\ -b_x b_z + n_i v_x v_z \\ \frac{n_n T_n}{m_n} + n_n u_x^2 \\ n_n u_x u_y \\ n_n u_x u_z \\ -\Phi \\ b_y v_x - b_x v_y \\ b_z v_x - b_x v_z \\ T_i v_x \\ T_n u_x \\ b_x c_h^2 \end{pmatrix}, \mathbf{F}_z = \begin{pmatrix} n_i v_z \\ n_n u_z \\ -b_x b_z + n_i v_x v_z \\ -b_y b_z + n_i v_y v_z \\ \frac{b_z}{2} - b_z^2 + n_i T_i + n_i v_z^2 \\ n_n u_x u_z \\ n_n u_y u_z \\ \frac{n_n T_n}{m_n} + n_n u_z^2 \\ -b_z v_x + b_x v_z \\ -b_z v_y + b_y v_z \\ -\Phi \\ T_i v_z \\ T_n u_z \\ b_z c_h^2 \end{pmatrix} \quad (16)$$

and the Jacobian matrices $\frac{d\mathbf{F}_x}{d\mathbf{u}}, \frac{d\mathbf{F}_z}{d\mathbf{u}}$ are given as

$$\frac{d\mathbf{F}_x}{d\mathbf{u}} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ T_i - v_x^2 & 0 & 2v_x & 0 & 0 & 0 & 0 & 0 & -b_x & b_y & b_z & n_i & 0 & 0 \\ -v_x v_y & 0 & v_y & v_x & 0 & 0 & 0 & 0 & -b_y & -b_x & 0 & 0 & 0 & 0 \\ -v_x v_z & 0 & v_z & 0 & v_x & 0 & 0 & 0 & -b_z & 0 & -b_x & 0 & 0 & 0 \\ 0 & \frac{T_n}{m_n} - u_x^2 & 0 & 0 & 0 & 2u_x & 0 & 0 & 0 & 0 & 0 & 0 & \frac{n_n}{m_n} & 0 \\ 0 & -u_x u_y & 0 & 0 & 0 & u_y & u_x & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -u_x u_z & 0 & 0 & 0 & u_z & 0 & u_x & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ \frac{-b_y v_x + b_x v_y}{n_i} & 0 & \frac{b_y}{n_i} & -\frac{b_x}{n_i} & 0 & 0 & 0 & 0 & -v_y & v_x & 0 & 0 & 0 & 0 \\ \frac{-b_z v_x + b_x v_z}{n_i} & 0 & \frac{b_z}{n_i} & 0 & -\frac{b_x}{n_i} & 0 & 0 & 0 & -v_z & 0 & v_x & 0 & 0 & 0 \\ -\frac{T_i v_x}{n_i} & 0 & \frac{T_i}{n_i} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & v_x & 0 & 0 \\ 0 & -\frac{T_n u_x}{n_n} & 0 & 0 & 0 & \frac{T_n}{n_n} & 0 & 0 & 0 & 0 & 0 & 0 & u_x & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_h^2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad (17)$$

$$\frac{d\mathbf{F}_z}{d\mathbf{u}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -v_x v_z & 0 & v_z & 0 & v_x & 0 & 0 & 0 & -b_z & 0 & -b_x & 0 & 0 & 0 \\ -v_y v_z & 0 & 0 & v_z & v_y & 0 & 0 & 0 & 0 & -b_z & -b_y & 0 & 0 & 0 \\ T_i - v_z^2 & 0 & 0 & 0 & 2v_z & 0 & 0 & 0 & b_x & b_y & -b_z & n_i & 0 & 0 \\ 0 & -u_x u_z & 0 & 0 & 0 & u_z & 0 & u_x & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -u_y u_z & 0 & 0 & 0 & 0 & u_z & u_y & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{T_n}{m_n} - u_z^2 & 0 & 0 & 0 & 0 & 0 & 2u_z & 0 & 0 & 0 & 0 & \frac{n_n}{m_n} & 0 \\ \frac{b_z v_x - b_x v_z}{n_i} & 0 & -\frac{b_z}{n_i} & 0 & \frac{b_x}{n_i} & 0 & 0 & 0 & v_z & 0 & -v_x & 0 & 0 & 0 \\ \frac{b_z v_y - b_y v_z}{n_i} & 0 & 0 & -\frac{b_z}{n_i} & \frac{b_y}{n_i} & 0 & 0 & 0 & 0 & v_z & -v_y & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ -\frac{T_i v_z}{n_i} & 0 & 0 & 0 & \frac{T_i}{n_i} & 0 & 0 & 0 & 0 & 0 & 0 & v_z & 0 & 0 \\ 0 & -\frac{T_n u_z}{n_n} & 0 & 0 & 0 & 0 & 0 & \frac{T_n}{n_n} & 0 & 0 & 0 & 0 & u_z & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & c_h^2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (18)$$

And the eigenvalues are

$$\lambda\left(\frac{d\mathbf{F}_x}{d\mathbf{u}}\right) = \begin{pmatrix} -iC_h \\ iC_h \\ u_x \\ u_x \\ u_x \\ -\sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_x \\ \sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_x \\ v_x \\ -\frac{b_x}{\sqrt{n_i}} + v_x \\ \frac{b_x}{\sqrt{n_i}} + v_x \\ -\frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i-\sqrt{b_x^4+2b_x^2(b_y^2+b_z^2-2n_iT_i)+(b_y^2+b_z^2+2n_iT_i)^2})}}{\sqrt{2n_i}} + v_x \\ \frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i-\sqrt{b_x^4+2b_x^2(b_y^2+b_z^2-2n_iT_i)+(b_y^2+b_z^2+2n_iT_i)^2})}}{\sqrt{2n_i}} + v_x \\ -\frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i+\sqrt{b_x^4+2b_x^2(b_y^2+b_z^2-2n_iT_i)+(b_y^2+b_z^2+2n_iT_i)^2})}}{\sqrt{2n_i}} + v_x \\ \frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i+\sqrt{b_x^4+2b_x^2(b_y^2+b_z^2-2n_iT_i)+(b_y^2+b_z^2+2n_iT_i)^2})}}{\sqrt{2n_i}} + v_x \end{pmatrix}, \quad (19)$$

$$\lambda\left(\frac{d\mathbf{F}_z}{d\mathbf{u}}\right) = \begin{pmatrix} -iC_h \\ iC_h \\ u_z \\ u_z \\ u_z \\ -\sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_z \\ \sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_z \\ v_z \\ -\frac{b_z}{\sqrt{n_i}} + v_z \\ \frac{b_z}{\sqrt{n_i}} + v_z \\ -\frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i-\sqrt{b_x^2+b_y^2+b_z^2+4n_iT_i(b_x^2+b_y^2-b_z^2)+4n_i^2T_i^2})}}{\sqrt{2n_i}} + v_z \\ \frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i-\sqrt{b_x^2+b_y^2+b_z^2+4n_iT_i(b_x^2+b_y^2-b_z^2)+4n_i^2T_i^2})}}{\sqrt{2n_i}} + v_z \\ -\frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i+\sqrt{b_x^2+b_y^2+b_z^2+4n_iT_i(b_x^2+b_y^2-b_z^2)+4n_i^2T_i^2})}}{\sqrt{2n_i}} + v_z \\ \frac{\sqrt{n_i(b_x^2+b_y^2+b_z^2+2n_iT_i+\sqrt{b_x^2+b_y^2+b_z^2+4n_iT_i(b_x^2+b_y^2-b_z^2)+4n_i^2T_i^2})}}{\sqrt{2n_i}} + v_z \end{pmatrix}, \quad (20)$$

which can be simplified as

$$\lambda\left(\frac{d\mathbf{F}_x}{d\mathbf{u}}\right) = \begin{pmatrix} -iC_h \\ iC_h \\ u_x \\ u_x \\ u_x \\ -\sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_x \\ \sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_x \\ v_x \\ -\frac{b_x}{\sqrt{n_i}} + v_x \\ \frac{b_x}{\sqrt{n_i}} + v_x \\ -c_{\text{fast},x}^- + v_x \\ c_{\text{fast},x}^- + v_x \\ -c_{\text{fast},x}^+ + v_x \\ c_{\text{fast},x}^+ + v_x \end{pmatrix}, \quad (21)$$

$$\lambda\left(\frac{d\mathbf{F}_z}{d\mathbf{u}}\right) = \begin{pmatrix} -iC_h \\ iC_h \\ u_z \\ u_z \\ u_z \\ -\sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_z \\ \sqrt{2}\sqrt{\frac{T_n}{m_n}} + u_z \\ v_z \\ -\frac{b_z}{\sqrt{n_i}} + v_z \\ \frac{b_z}{\sqrt{n_i}} + v_z \\ -c_{\text{fast},z}^- + v_z \\ c_{\text{fast},z}^- + v_z \\ -c_{\text{fast},z}^+ + v_z \\ c_{\text{fast},z}^+ + v_z \end{pmatrix}, \quad (22)$$

where

$$c_{\text{fast},x}^+ = \frac{1}{\sqrt{2}}\sqrt{c_A^2 + \frac{2}{\gamma}c_s^2 + \sqrt{(c_A^2 + \frac{2}{\gamma}c_s^2)^2 - \frac{8}{\gamma}c_{A,x}^2c_s^2}}, \quad (23)$$

$$c_{\text{fast},z}^+ = \frac{1}{\sqrt{2}}\sqrt{c_A^2 + \frac{2}{\gamma}c_s^2 + \sqrt{(c_A^2 + \frac{2}{\gamma}c_s^2)^2 - \frac{8}{\gamma}c_{A,z}^2c_s^2}}, \quad (24)$$

$$c_{\text{fast},x}^- = \frac{1}{\sqrt{2}}\sqrt{c_A^2 + \frac{2}{\gamma}c_s^2 - \sqrt{(c_A^2 + \frac{2}{\gamma}c_s^2)^2 - \frac{8}{\gamma}c_{A,x}^2c_s^2}}, \quad (25)$$

$$c_{\text{fast},z}^- = \frac{1}{\sqrt{2}}\sqrt{c_A^2 + \frac{2}{\gamma}c_s^2 - \sqrt{(c_A^2 + \frac{2}{\gamma}c_s^2)^2 - \frac{8}{\gamma}c_{A,z}^2c_s^2}}, \quad (26)$$

and

$$c_A = \sqrt{\frac{b_x^2 + b_y^2 + b_z^2}{n_i}}, \quad c_{A,x} = \frac{b_x}{\sqrt{n_i}}, \quad c_{A,z} = \frac{b_z}{\sqrt{n_i}}, \quad c_s = \sqrt{\gamma T_i}. \quad (27)$$

2.7 Second Order Accuracy in Time

Here U denotes conservative variables and u primitive variables, and $f(u)$ and $f(U)$ both stand for the flux function, because it can be calculated in both ways. First we calculate $\tilde{U}^{n+1} = U^n - \frac{\Delta t}{\Delta x}[f(u_{i+\frac{1}{2}}^l) -$

$f(u_{i-\frac{1}{2}}^r)]$ as well as $\tilde{u}^{n+1} = u(\tilde{U}^{n+1})$, and then 2nd order reconstruction is

$$\begin{aligned} u_{i+\frac{1}{2}}^L &= \frac{1}{2}(u_i^n + \tilde{u}_i^{n+1}) + \phi(r_i)(u_{i+1}^n - u_i^n) \\ u_{i+\frac{1}{2}}^R &= \frac{1}{2}(u_{i+1}^n + \tilde{u}_{i+1}^{n+1}) - \phi(r_{i+1})(u_{i+2}^n - u_{i+1}^n). \end{aligned} \quad (28)$$

The numerical fluxes are obtained by

$$F_{i+\frac{1}{2}} = \frac{1}{2}([f(u_{i+\frac{1}{2}}^L) + f(u_{i+\frac{1}{2}}^R)] - \lambda_{i+1/2}[U(u_{i+\frac{1}{2}}^R) - U(u_{i+\frac{1}{2}}^L)]). \quad (29)$$

In this way, even though we, as you can see next, adopt forward Euler time stepping, the scheme is still of 2nd accuracy in time. Primitive variables are used for reconstruction, as we want to eliminate their errors.

2.8 Semi-implicit Time Integration

A semi-implicit Euler time-stepping is adopted, because it's said the general Lagrange multiplier method is unstable in other semi-implicit methods (have not done the analysis yet).

The expression is

$$u^{n+1} = u^n + (R_E^n + R_I^{n+1})\Delta t. \quad (30)$$

Here R_I are the implicit terms and consists of the collisional and diffusive terms, like collision rate, resistivity, and heat flux. R_E contains the explicit terms, which are the source terms (for the chromosphere model, the sources are not considered yet), and the numerical fluxes calculated as shown above.

We can use an iteration method to obtain $R_I^{n+1} = R_I(u^{n+1})$. Here are the steps:

1. $\tilde{u}^{n+1} = u^n + R_E^n \Delta t$
2. $\tilde{R}_I^{n+1} = R_I(\tilde{u}^{n+1})$
3. $u^{n+1} = u^n + (\tilde{R}_I^{n+1} + R_E^n) \Delta t$
4. In a new iteration, $\tilde{u}^{n+1}$ is set equal to u^{n+1} . Repeat step 2 and 3 until $|\tilde{u}^{n+1} - u^n| < \epsilon$, where ϵ is the tolerance.

3 Field-Aligned Motions

Assuming the plasma flow is parallel or anti-parallel to the magnetic field and $\mathbf{V} = V\mathbf{b}$, $\mathbf{U} = U\mathbf{b}$, and substituting T equations with energy equations where $e = \frac{3}{2}p + \frac{1}{2}\rho v^2 + \rho\phi_g$, where $\nabla\phi_g = -\mathbf{g}$ we obtain

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\rho_i V \mathbf{b}) = 0 \quad (31)$$

$$\frac{\partial \rho_n}{\partial t} + \nabla \cdot (\rho_n U \mathbf{b}) = 0 \quad (32)$$

$$\frac{\partial \rho_i V}{\partial t} + \nabla \cdot [\rho_i V^2 + n_i k_b T_i] \mathbf{b} = (n_i k_b T_i) \nabla \cdot \mathbf{b} - \rho_i (\mathbf{b} \cdot \nabla) \phi_g - \rho_i \nu_{in} (V - U) \quad (33)$$

$$\frac{\partial \rho_n U}{\partial t} + \nabla \cdot [\rho_n U^2 + n_n k_b T_n] \mathbf{b} = (n_n k_b T_n) \nabla \cdot \mathbf{b} - \rho_n (\mathbf{b} \cdot \nabla) \phi_g + \rho_i \nu_{in} (V - U) \quad (34)$$

$$\frac{\partial e_i}{\partial t} + \nabla \cdot [(e_i + n_i k_b T_i) V \mathbf{b}] = -\nabla \cdot \mathbf{q}_i + \frac{m_i}{m_i + m_n} \nu_{in} n_i [3k_b (T_n - T_i) + m_n |V - U|^2] - \rho_i \nu_{in} V (V - U), \quad (35)$$

$$\frac{\partial e_n}{\partial t} + \nabla \cdot [(e_n + n_n k_b T_n) U \mathbf{b}] = -\nabla \cdot \mathbf{q}_n + \frac{m_n}{m_i + m_n} \nu_{in} n_n [3k_b (T_i - T_n) + m_i |V - U|^2] + \rho_i \nu_{in} U (V - U), \quad (36)$$

$$\mathbf{q} = -\kappa \nabla T, \quad (37)$$

where the form of $\mathbf{q}$ is uncertain. It can also be, as is suggested by Xianyu, $\mathbf{q} = -\kappa \mathbf{b} \mathbf{b} \cdot \nabla T$.

The equations, however, exhibit certain limitations. Specifically, electron pressure and heat conduction are neglected, despite the assumption that $T_e = T_i$. By incorporating quasi-neutrality $n_i = n_e$, these effects can be accounted for in the equation of motion and the equation of energy by

$$p_i = n_i k_b T_i + n_e k_b T_e = 2n_i k_b T_i, \quad (38)$$

and

$$\mathbf{q}_e = -\kappa_e T_e^{5/2} \mathbf{b} \mathbf{b} \cdot \nabla T_e, \quad \kappa_e \approx 9.2 \times 10^{-12} \text{ W m}^{-1} \text{ K}^{-7/2} \quad (39)$$

The equations thus become

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\rho_i V \mathbf{b}) = 0 \quad (40)$$

$$\frac{\partial \rho_n}{\partial t} + \nabla \cdot (\rho_n U \mathbf{b}) = 0 \quad (41)$$

$$\frac{\partial \rho_i V}{\partial t} + \nabla \cdot [\rho_i V^2 + 2n_i k_b T_i] \mathbf{b} = (2n_i k_b T_i) \nabla \cdot \mathbf{b} - \rho_i (\mathbf{b} \cdot \nabla) \phi_g - \rho_i \nu_{in} (V - U) \quad (42)$$

$$\frac{\partial \rho_n U}{\partial t} + \nabla \cdot [\rho_n U^2 + n_n k_b T_n] \mathbf{b} = (n_n k_b T_n) \nabla \cdot \mathbf{b} - \rho_n (\mathbf{b} \cdot \nabla) \phi_g + \rho_i \nu_{in} (V - U) \quad (43)$$

$$\frac{\partial e_i}{\partial t} + \nabla \cdot [(e_i + 2n_i k_b T_i) V \mathbf{b}] = -\nabla \cdot \mathbf{q}_i - \nabla \cdot \mathbf{q}_e + \frac{m_i}{m_i + m_n} \nu_{in} n_i [3k_b (T_n - T_i) + m_n |V - U|^2] - \rho_i \nu_{in} V (V - U), \quad (44)$$

$$\frac{\partial e_n}{\partial t} + \nabla \cdot [(e_n + n_n k_b T_n) U \mathbf{b}] = -\nabla \cdot \mathbf{q}_n + \frac{m_n}{m_i + m_n} \nu_{in} n_n [3k_b (T_i - T_n) + m_i |V - U|^2] + \rho_i \nu_{in} U (V - U). \quad (45)$$

Here, ρ_i , V , and e_i represent quantities associated with charged particles, including both ions and electrons, while n_i and T_i are specific to ions. Since ions dominate these quantities, ρ_i , V , and e_i can be reasonably approximated as ion-related. For instance, $\rho_i \approx m_i n_i$, where m_i is the ion mass.

Because electrons move significantly faster than ions and neutrals, the electron heat conduction dominates over ion and neutral heat conduction. However, in regions with low ionization degrees, neutral heat conduction can dominate.

3.1 Electron Heat Conductivity

In this subsection, we analyze the electron heat conductivity in a partially ionized plasma by incorporating the effect of electron-neutral collision frequency.

We begin with Spitzer's collisional formulation:

$$\kappa_e = (4\pi\epsilon_0)^2 \frac{20m_e^2 k_B v_{th}^5}{3e^4 Z \ln \Lambda} \left(\frac{2}{3\pi} \right)^{\frac{3}{2}} \delta_T \epsilon, \quad (46)$$

where ϵ_0 , m_e , k_B , v_{th} , and e represent the vacuum permittivity, electron mass, Boltzmann constant, root-mean-square velocity, and electron charge, respectively. The parameters ϵ and δ_T are dimensionless coefficients. For $Z = 1$, these coefficients take the values:

$$\delta_T = 0.2252, \quad \epsilon = 0.4189. \quad (47)$$

Next, we account for the contribution from electron-neutral collisions by considering a more general form of thermal conductivity in fluids, which can be expressed as:

$$\kappa \propto k_B n_e v_{th} \lambda_{\text{mfp}} \sim \frac{k_B n_e v_{th}^2}{\nu} \sim \frac{k_B^2 n_e T_e}{m_e \nu}, \quad (48)$$

where λ_{mfp} and ν are the mean free path between collisions and the collision frequency, respectively.

In Spitzer's formulation, both electron-ion and electron-electron collisions are considered, with the respective collision frequencies approximated as:

$$\nu_{ee} \approx \nu_{ei} = \frac{n_e e^4}{4\pi\epsilon_0^2 \sqrt{k_B^2 m_e T_e^3}} \ln \Lambda. \quad (49)$$

Since $\nu = \nu_{ei} + \nu_{ee} \approx 2\nu_{ei}$, we can rewrite the electron heat conductivity κ_e in a more familiar form:

$$\kappa_e = 320 \sqrt{\frac{2}{\pi}} \delta_T \epsilon \frac{k_B^2 n_e T_e}{m_e \nu}. \quad (50)$$

To include the effect of electron-neutral collisions, we now substitute ν with the total collision frequency $\nu_{\text{tot}} = \nu_{ei} + \nu_{ee} + \nu_{en}$, where ν_{en} is the electron-neutral collision frequency. It can be expressed as:

$$\nu_{en} = n_n \sigma_{\text{en}} \int v f_e(v) dv = n_n \sigma_{\text{en}} \sqrt{\frac{8k_B T_e}{\pi m_e}} = n_n r_n^2 \sqrt{\frac{8\pi k_B T_e}{m_e}} = 5.5 \times 10^{-17} n_n \sqrt{T_e} \quad (\text{in SI units}), \quad (51)$$

where $\sigma_{\text{en}} \approx \pi r_n^2$ is the electron-neutral cross-section, r_n is the neutral particle radius, and $f_e(v)$ is the Maxwellian velocity distribution for electrons. However, this value may not be accurate. Instead, $\nu_{en} = 1.95 \times 10^{-16} n_n \sqrt{T_e}$ will be employed.

Finally, the modified expression for the electron heat conductivity κ_e becomes:

$$\kappa_e = 320 \sqrt{\frac{2}{\pi}} \delta_T \epsilon \frac{k_B^2 n_e T_e}{m_e \nu_{\text{tot}}}. \quad (52)$$

In SI units, this can be expressed as

$$\kappa_e = \frac{9.2048 \times 10^{-12} n_e T_e^{5/2}}{n_e + 3.5609 \times 10^{-12} n_n T_e^2}. \quad (53)$$

If we take $T_e = T_n = 6500K$, we obtain

$$\kappa_e = \frac{0.0313545 n_e}{n_e + 0.00015045 n_n}. \quad (54)$$

It can be concluded that only when $n_n \sim 1000 n_e$ do electron-neutral collisions contribute significantly to the electron heat conductivity. In the case where $n_n \sim n_e$, $\kappa_e \approx 0.031 \text{ W/(m}\cdot\text{K)}$.

3.2 Neutral Heat Conduction

The thermal conductivity of a neutral gas can be expressed as

$$\kappa_n = \frac{1}{3} c_v \rho \bar{c} \lambda_{\text{mfp}}, \quad (55)$$

where c_v , ρ , $\bar{c}$, and λ_{mfp} represent the heat capacity per unit mass, mass density, mean velocity, and mean free path, respectively.

For H gas, we have $\rho c_v = \left(\frac{\partial u}{\partial T}\right)_V = \frac{3}{2} n_n k_B$. Thus, κ_n takes the form:

$$\kappa_n = \frac{1}{2} n_n k_B \frac{\bar{c}^2}{\nu} = \frac{4}{\pi} \frac{k_B^2 n_n T_n}{m_n \nu}, \quad (56)$$

where $\nu = \nu_{ni} + \nu_{nn}$, and $\bar{c}$ is substituted by $\sqrt{\frac{8k_B T_n}{\pi m_n}}$ based on the Maxwellian velocity distribution.

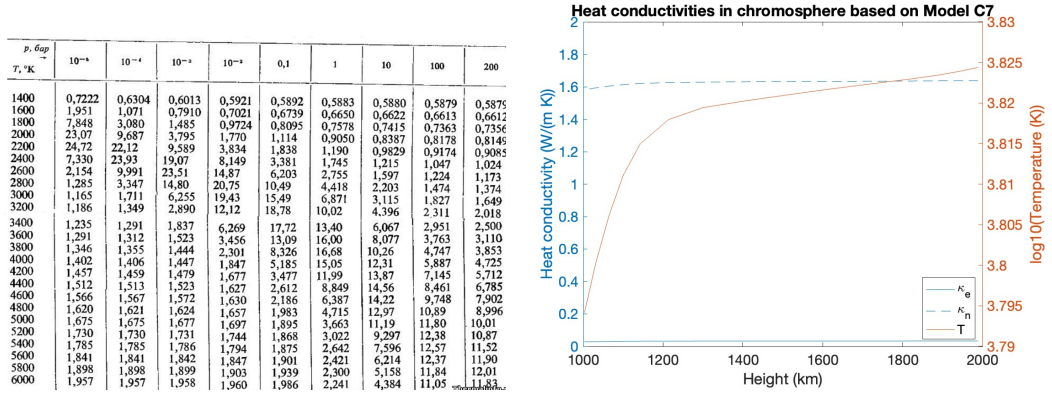


Figure 1: A comparison between experimental hydrogen thermal conductivity and the one obtained from (59). From the table it can be extrapolated that in the chromosphere where $T \approx 6000K$ and $n \approx 2 \times 10^{16}m^{-3}$, the hydrogen gas, which is mostly composed of atomic hydrogen, has the thermal conductivity around 2 W/(m K). On the other hand, if we neglect ions, the calculation from (59) will give approximately 1.6 W/(m K).

When approximating $m_i \approx m_n$, the collision frequencies ν_{ni} and ν_{nn} can be derived from the Maxwellian distribution of relative velocities. The relative velocity between ions and neutrals is characterized by $v_{ni}^{\text{rel}} \sim f(T_i + T_n)$, and between neutrals by $v_{nn}^{\text{rel}} \sim f(2T_n)$, yielding:

$$\nu_{ni} = n_i \sigma_{ni} \sqrt{\frac{8k_B(T_i + T_n)}{\pi m_n}} = n_i (r_i + r_n)^2 \sqrt{\frac{8\pi k_B(T_i + T_n)}{m_n}}, \quad (57)$$

$$\nu_{nn} = 4n_n \sigma_{nn} \sqrt{\frac{k_B T_n}{\pi m_n}} = 4n_n (2r_n)^2 \sqrt{\frac{\pi k_B T_n}{m_n}}. \quad (58)$$

κ_n , then, can be expressed in SI units as

$$\kappa_n = \frac{0.0342006 n_n T_n}{1.20613 n_i \sqrt{T_n + T_i} + 1.70573 n_n \sqrt{T_n}}. \quad (59)$$

If we take $T_n = T_i = 6500K$, $n_i = n_n$, we obtain

$$\kappa_n = 0.947 \text{ W/(m} \cdot \text{K)} \quad (60)$$

Note that this value is far greater than κ_e at the same temperature. It is only when the temperature exceeds 100,000K that $\kappa_n = 3.71 \text{ W/(m} \cdot \text{K)}$ becomes one order of magnitude smaller than $\kappa_e = 29.1 \text{ W/(m} \cdot \text{K)}$.

3.3 Validation and Application to Model C7

A comparison between experimental hydrogen thermal conductivity and the one obtained from (59) is shown in Figure 1.

Then we apply the heat conductivity formulation to Model C7, where the plasma profile such as ion density, total hydrogen density, atomic hydrogen density and the temperature, is given (see Figure 2). It can be seen that neutral heat conductivity dominates over electron in the chromosphere. However, the role of two heat conductivities switches in transition region where the temperature ramps up.

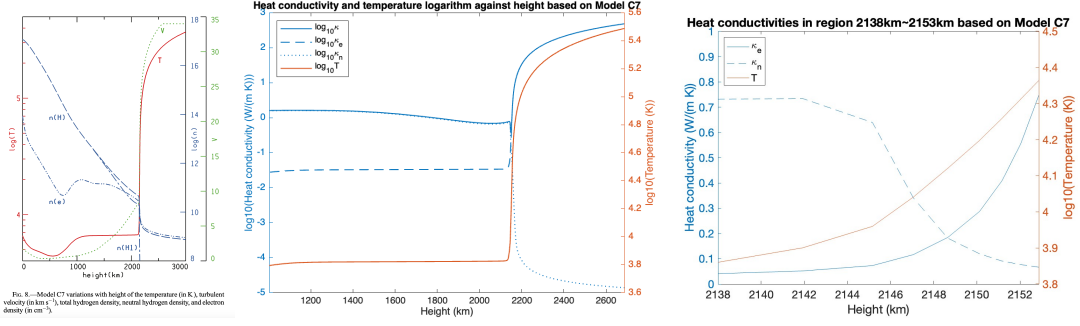


Figure 2: **Left:** Model C7 profile including ion density, total hydrogen density, atomic hydrogen density and the temperature. **Middle:** heat conductivity logarithms and temperature logarithm against height calculated from the profile. **Right:** heat conductivities and temperature logarithms against height (zoomed in). It can be seen that neutral heat conductivity dominates over electron in the chromosphere. However, the role of two heat conductivities switches in transition region where the temperature ramps up. The intersection of two conductivities occurs at height=2149km.

3.4 Fitted Model

We evaluate the sum of two heat conductivities, $\kappa = \kappa_n + \kappa_e$, where κ can also be expressed as a function of temperature in Model C7 due to the monotonic relationship between temperature and height. This relationship is illustrated in Figure 3. A notable drop in conductivity is observed in the temperature range [6400 K, 6700 K], primarily caused by changes in ionization levels in the chromosphere, where the temperature variation is relatively small compared to the ionization variation.

The total conductivity $\kappa(T)$ can be divided into two distinct regions: a low-temperature region ($T < 2.5 \times 10^4$ K) dominated by κ_n , and a high-temperature region ($T > 2.5 \times 10^4$ K) dominated by κ_e . For the fitted curve, a constant value of 0.89 W/(m·K) represents the average conductivity in the low-temperature region, while $\kappa_e^{\text{Spitzer}}$ is used for the high-temperature region. The critical temperature is chosen to ensure that the fitted curve, as a piecewise function, remains continuous across both regions.

Thus, the fitted conductivity $\kappa^{\text{fitted}}(T)$ can be expressed as a piecewise function in SI units:

$$\kappa^{\text{fitted}}(T) = \begin{cases} 0.89 \text{ W/(m·K)}, & \text{if } T < 2.5 \times 10^4 \text{ K} \\ 9.2 \times 10^{-12} T^{5/2} \text{ W/(m·K)}, & \text{if } T \geq 2.5 \times 10^4 \text{ K} \end{cases}$$

This piecewise representation ensures a smooth transition between the two temperature regions.

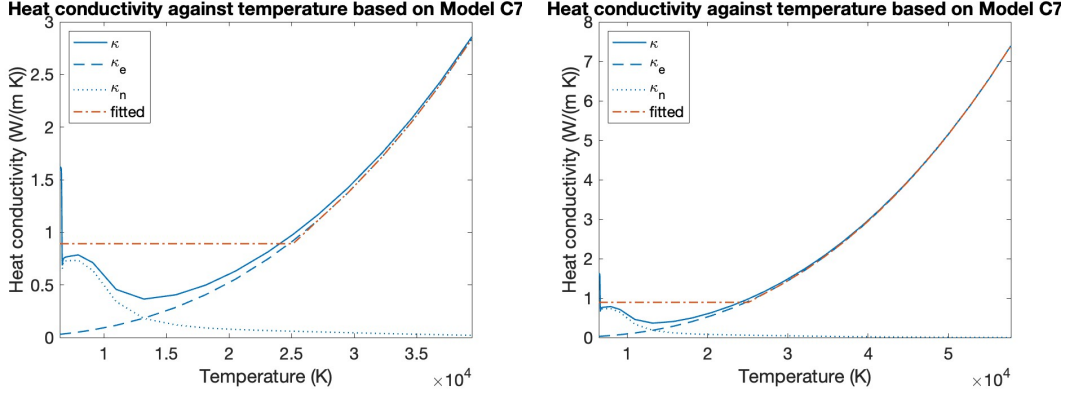


Figure 3: The heat conductivities κ_e , κ_n , and their sum $\kappa = \kappa_e + \kappa_n$ as functions of temperature, based on Model C7, are shown here (**left**: zoomed in; **right**: zoomed out). The total conductivity $\kappa(T)$ can be divided into two distinct regions: a low-temperature region ($T < 2.5 \times 10^4$ K), where κ_n dominates, and a high-temperature region ($T > 2.5 \times 10^4$ K), where κ_e is dominant. For the fitted curve, a constant value of 0.89 W/(m-K) is used to represent the average conductivity in the low-temperature region, while $\kappa_e^{\text{Spitzer}}$ is applied for the high-temperature region. The critical temperature is selected such that the fitted curve, as a piecewise function, remains continuous across both regions.

3.5 Vector Form

The MHD equations can then be rewritten in the form of derivatives along the field line as

$$\frac{\partial}{\partial t} \begin{bmatrix} \rho_i \\ \rho_n \\ \rho_i V \\ \rho_n U \\ e_i \\ e_n \end{bmatrix} + B \frac{\partial}{\partial s} \begin{bmatrix} \frac{\rho_i V}{B} \\ \frac{\rho_n U}{B} \\ \frac{\rho_i V^2 + 2n_i k_b T_i}{B} \\ \frac{\rho_n U^2 + n_n k_b T_n}{B} \\ \frac{(e_i + 2n_i k_b T_i) V}{B} \\ \frac{(e_n + n_n k_b T_n) U}{B} \end{bmatrix} = \mathbf{S}, \quad (61)$$

where

$$\mathbf{S} = \begin{bmatrix} 0 \\ 0 \\ 2n_i k_b T_i B \frac{\partial}{\partial s} \left(\frac{1}{B} \right) - \rho_i \frac{\partial}{\partial s} \phi_g - \rho_i \nu_{in} (V - U) \\ n_n k_b T_n B \frac{\partial}{\partial s} \left(\frac{1}{B} \right) - \rho_n \frac{\partial}{\partial s} \phi_g + \rho_i \nu_{in} (V - U) \\ \left[(\kappa_i + \kappa_e) B \frac{\partial}{\partial s} \left(\frac{1}{B} \right) + \frac{\partial}{\partial s} (\kappa_i + \kappa_e) \right] \frac{\partial}{\partial s} T_i + \frac{1}{2} \nu_{in} n_i [3k_b (T_n - T_i) + m_n |V - U|^2] - \nu_{in} \rho_i V (V - U) \\ \left[\kappa_n B \frac{\partial}{\partial s} \left(\frac{1}{B} \right) + \frac{\partial}{\partial s} \kappa_n \right] \frac{\partial}{\partial s} T_n + \frac{1}{2} \nu_{in} n_n [3k_b (T_i - T_n) + m_i |V - U|^2] + \nu_{in} \rho_i U (V - U) \end{bmatrix} \quad (62)$$

i.e.

$$\frac{\partial}{\partial t} \left(\frac{\mathbf{U}}{B} \right) + \frac{\partial}{\partial s} \left(\frac{\mathbf{F}}{B} \right) = \frac{\mathbf{S}}{B} \quad (63)$$

(63) has the same form as $\frac{\partial}{\partial t} \mathbf{U}' + \frac{\partial}{\partial s} \mathbf{F}' = \mathbf{S}'$ and thus, in the next subsection, we can obtain the eigen system similarly.

3.6 Eigen System

The eigenvalues are

$$\begin{pmatrix} u \\ v \\ v - \sqrt{2\gamma k_b n_i T_i / \rho_i} \\ v + \sqrt{2\gamma k_b n_i T_i / \rho_i} \\ u - \sqrt{\gamma k_b n_n T_n / \rho_n} \\ u + \sqrt{\gamma k_b n_n T_n / \rho_n} \end{pmatrix} \quad (64)$$

where $\gamma = \frac{5}{3}$. It can be seen that without the induction equation, c_{fast} is degraded into $c_s = \sqrt{\gamma p / \rho}$. The spectral radius of the Jacobian can thus be expressed as

$$\rho \left(\frac{\partial \mathbf{F}'}{\partial \mathbf{U}'} \right) = \max(|v| + |c_{s,i}|, |u| + |c_{s,n}|) \quad (65)$$

3.7 Numerical Methodology

$$\frac{\partial \mathbf{U}_i}{\partial t} = -\frac{B_i}{ds_i} \left(\frac{\mathbf{F}_{i+\frac{1}{2}}}{B_{i+\frac{1}{2}}} - \frac{\mathbf{F}_{i-\frac{1}{2}}}{B_{i-\frac{1}{2}}} \right) + \mathbf{S}_i \quad (66)$$

4 Numerical Results

The initial conditions are defined by Model C7, as listed in Table 1. A straight field line comprising 100 cells is generated using spline interpolation, each cell having an equal length. The magnetic field B is held constant at 1 T and gravitational forces are not yet taken into account.

For boundary conditions, the inner boundaries remain fixed at the initial values, while the outer boundaries are adjusted: the temperature at the outermost boundary is set to twice the initial temperature of the outermost cell, and the velocity at each time step is updated to be half of the outermost cell's velocity.

Height (km)	Electron Density (m ⁻³)	H Atom Density (m ⁻³)	Temperature (K)
1.003e+03	1.903e+17	2.693e+19	6.225e+03
1.032e+03	2.021e+17	2.179e+19	6.315e+03
1.065e+03	2.091e+17	1.722e+19	6.400e+03
1.101e+03	2.104e+17	1.340e+19	6.474e+03
1.143e+03	1.995e+17	1.009e+19	6.531e+03
1.214e+03	1.760e+17	6.322e+18	6.576e+03
1.299e+03	1.489e+17	3.641e+18	6.598e+03
1.398e+03	1.438e+17	1.880e+18	6.610e+03
1.520e+03	1.423e+17	7.956e+17	6.623e+03
1.617e+03	1.267e+17	4.012e+17	6.633e+03
1.722e+03	1.027e+17	1.916e+17	6.643e+03
1.820e+03	7.969e+16	9.843e+16	6.652e+03
1.894e+03	6.450e+16	6.005e+16	6.660e+03
1.946e+03	5.526e+16	4.371e+16	6.667e+03
1.989e+03	4.826e+16	3.447e+16	6.674e+03

Table 1: Height, Electron Number Density, H Atom Number Density, and Temperature

The ion sound speed C_{si} for this profile is approximately 1.3×10^4 m/s. To reach a steady-state solution, the total simulation time is set to $100L/C_{si}$, where L is the length of the field line, i.e. 0.986×10^3 kmx.